

Week 3: Multi-Agent Systems and Emergent Behavior

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Environment Setup & Mathematical Model

The simulation operates as a decentralized multi-agent system where complex global behavior emerges strictly from local interactions. The environment is a 2D continuous space.

Each agent (boid) i is defined by its position vector $\vec{p}_i$ and velocity vector $\vec{v}_i$. There is no central controller; instead, at every discrete time step, each agent independently calculates a new acceleration vector based on the states of neighboring agents j that fall within its local perception radius ($\|\vec{p}_i - \vec{p}_j\| < R$).

The navigation of each agent is governed by a weighted sum of three primary steering forces:

1. Separation

This force steers the boid away from local flockmates to prevent collisions. The repulsion is calculated as the sum of vectors pointing away from neighbors, weighted inversely by distance to prioritize immediate threats.

$$\vec{F}_{sep} = \sum_{j \neq i} \frac{\vec{p}_i - \vec{p}_j}{\|\vec{p}_i - \vec{p}_j\|^2} \quad (1)$$

2. Alignment

This force steers the boid towards the average heading of its local flockmates, keeping the group moving in a unified direction.

$$\vec{F}_{ali} = \left(\frac{1}{N} \sum_{j \neq i} \vec{v}_j \right) - \vec{v}_i \quad (2)$$

3. Cohesion

This force steers the boid towards the average position (center of mass) of its local flockmates, ensuring the group remains clumped together rather than drifting apart.

$$\vec{F}_{coh} = \left(\frac{1}{N} \sum_{j \neq i} \vec{p}_j \right) - \vec{p}_i \quad (3)$$

The final acceleration applied to agent i is scaled by user-defined weights for each rule, yielding organic flocking dynamics.

1 Task 2: Parameter Exploration

1.1 Stable Flocking

Parameters: Radius = 50, Alignment = 2.0, Separation = 1.0, Cohesion = 1.0

Observations: The flock achieves a dynamic equilibrium. Agents group together and move in a unified direction without colliding.

1.2 Fragmentation

Parameters: Radius = 10, Alignment = 2.0, Separation = 1.5, Cohesion = 1.0

Observations: The reduced perception radius and Strong alignment cause the flock to break into several smaller, disconnected groups.

1.3 Oscillation / Chaos

Parameters: Radius = 10, Alignment = 0.5, Separation = 2.0, Cohesion = 0.1

Observations: The overwhelming separation and lack of cohesion and alignment cause agents to clump too tightly, resulting in jerky, oscillating movements as they constantly overshoot the center of mass.

Increasing the Perception radius does not decrease the local chaos but overall it seems like the flock is moving in one direction.

Task 1 Videos: [Click here to watch the Stable Flocking, Fragmentation and Chaos Videos.](#)

2 Task 4: Predator and Prey Observations

Group Formation: Before a predator approaches, the prey agents maintain standard cohesion.

Escape Behavior & Splitting: When a predator breaches the perception threshold, the flee force overwhelms cohesion. The flock violently fragments as individuals prioritize immediate survival over group unity.

Merging: Once the predator passes and is no longer detected locally, the baseline cohesion and alignment rules naturally pull the scattered prey back into a unified flock.

Task 4 Videos: [Click here to watch the Prey and Predator Observations.](#)

3 Task 5: Information Spread Observations

The simulation initializes with a single "informed" agent (e.g., carrying an evacuation signal). When uninformed agents enter the transmission radius of an informed agent, the state transfers. This creates a cascading, non-linear spread of information resembling a viral contagion or rumor spread, ultimately resulting in the entire population moving toward the designated safe zone.

Task 5 Video: [Click here to watch the Information Spread observations.](#)

4 Reflection Questions

1. **Why does flocking emerge?** Flocking emerges because independent entities simultaneously applying local repulsive, alignment, and attractive forces naturally find a stable spatial equilibrium.
 2. **Why is no leader required?** The system relies on decentralized, local information. Global coordination is a byproduct of overlapping local neighborhoods rather than top-down instruction.
 3. **What happens when perception radius becomes very small?** Agents cannot detect enough neighbors to form a consensus, leading to fragmentation and independent, random wandering.
 4. **What happens when perception radius becomes very large?** The simulation approaches a centralized state where every agent considers the entire population, leading to a single, monolithic, and less fluid clump.
 5. **What causes a flock to fragment?** High separation weights, low cohesion, or the introduction of a disruptive external force (like a predator or an obstacle).
 6. **How is Boids similar to Cellular Automata?** Both generate complex global patterns from simple, local, neighbor-based rules evaluated at discrete time steps.
 7. **How is Boids different from Neural Cellular Automata?** Boids operate in continuous space with continuous velocity vectors and hardcoded physics rules, whereas NCAs operate on a discrete grid using learned, distributed neural network parameters.
 8. **What examples of emergent behavior exist in the real world?** Traffic jams, stock market fluctuations, immune system responses, and the formation of paths in parks (desire paths).
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